

Deliverable 3: Use-Case Flow Specification

Problem Statement #02 | Automated Rubric Assignment Evaluator

Use Case ID	UC-01
Use Case Name	Submit & Auto-Evaluate Project
Primary Actor	Student
Supporting Actors	Sandboxed Test Execution Engine, Notification Service
Trigger	The student uploads a project archive before the deadline.
Related Requirements	FR-001, FR-003, NFR-001, NFR-002
Included Use Cases	UC-02 Validate Submission Package, UC-03 Execute Automated Test Suite, UC-04 Generate Rubric Score Breakdown
Extension Point	UC-08 Override Rubric Score, when the evaluator disputes the automatic total

Preconditions

- P1** The student is logged in and is enrolled in the course the assignment belongs to.
- P2** A rubric version is live and tied to a configured unit test suite (FR-002), and the test execution engine is up.
- P3** The submission window is still open and the student has attempts left.

Postconditions

Success guarantee: The archive is saved with its own submission ID and a timestamp. An itemised rubric breakdown is stored against that ID and shown to both the student and the faculty evaluator. The status is set to Evaluated and the submission goes into the pool for peer review (UC-06).

Minimal guarantee: If the evaluation cannot finish, no score is published at all. The submission is kept with a failure status and a reason code, and the student can upload again while the window is open.

Main Success Scenario

- Student picks the assignment and chooses "Upload Submission". The system shows the upload form along with the rules for the archive: allowed format, 25 MB limit, and the folder structure it expects.
- Student picks a project archive and confirms the upload.
- System takes the archive and creates a submission record holding a unique ID, the owner and a server timestamp.
- System checks the archive against the format, size and structure rules «include» UC-02.
- System puts the checked submission on the evaluation queue and shows the status as Queued, with its place in line.
- System sends the submission to the sandboxed test execution engine. The engine runs the configured unit test suite with a 60 second CPU limit and no network access «include» UC-03.
- The engine returns a result for every test case, giving its identifier, whether it passed or failed, and how long it took.
- System matches each test case result to its rubric criterion and works out an itemised breakdown against the rubric version that is live «include» UC-04.
- System saves the breakdown, notes which rubric version was used, and sets the status to Evaluated.
- System shows the student the breakdown, listing the marks for each criterion, the test cases that failed, and the total. The faculty evaluator is notified through the notification service «include» UC-09.
- System adds the submission to the pool for peer review. The use case ends here.

Alternate & Exception Flows

4a. Archive fails validation

- System turns the archive away before it reaches the queue and says which rule was broken, for example ".rar is not a permitted format" or "required src/ directory not found".
- System throws the upload away and does not count it as an attempt, sending the student back to step 1. If a corrected archive is uploaded the flow picks up again at step 2. If not, the use case ends with no score.

6a. Sandbox policy violation (CPU limit or attempted network access)

- The engine kills the offending process and returns the reason code POLICY_VIOLATION. Nothing else on the queue is affected (NFR-002).
- System sets the status to Execution Halted (Policy Violation), gives zero for every criterion that relied on the run that was killed, and saves the reason against the submission.
- System tells both the student and the evaluator, marks the submission for manual review, and carries on from step 9 with the partial breakdown.

6b. Test Execution Engine unavailable

- System keeps the submission at Queued and tries to send it three more times, waiting longer between each try. If all three fail the status becomes Evaluation Pending and the faculty evaluator is alerted. The submission goes back on the queue by itself once the engine is working again. Nothing is thrown away (NFR-001).

10a. Faculty evaluator disputes an automatic criterion score «extend» UC-08

- Evaluator changes the score and writes a reason for it. The system keeps a permanent record of the old and new values and publishes the corrected breakdown (FR-005).